

DEMO-301:2026

Illustrative High Strength Deformed Steel Bars for Concrete Reinforcement Specification (synthetic demo document, not an Indian Standard)

Fourth Revision - ICS 77.140.15

FOREWORD

This is a synthetic demonstration document created for the BISense prototype. It is not an Indian Standard, it has not been issued by the Bureau of Indian Standards, and all values in it are illustrative only.

This illustrative specification covers deformed steel bars used as reinforcement in concrete, commonly called TMT bars when produced by thermo-mechanical treatment.

1 SCOPE

1.1 This document covers the requirements of deformed steel bars and wires for use as reinforcement in concrete, in grades DFe 415, DFe 500 and DFe 550.

2 REFERENCES

The following documents contain provisions which, through reference in this text, constitute provisions of this document:

DEMO-900 Illustrative Rules for Rounding Off Numerical Values

DEMO-920 Illustrative Method for Tensile Testing of Steel Products

3 TERMINOLOGY

For the purpose of this document, the following definitions shall apply.

3.1 Deformed Bar - A bar with ribs or other deformations on its surface that increase its bond with concrete.

3.2 Nominal Diameter - The diameter of a plain round bar having the same mass per metre length as the deformed bar.

3.3 Thermo-Mechanical Treatment - A process of quenching a hot-rolled bar with water followed by self-tempering, giving a tough core and a hard surface.

4 REQUIREMENTS

4.1 Chemical Composition

The ladle analysis of steel shall not exceed the limits given in Table 1.

Table 1 Chemical Composition

Constituent	DFe 415, Max percent	DFe 500, Max percent	DFe 550, Max percent
Carbon	0.30	0.30	0.30
Sulphur	0.060	0.055	0.055
Phosphorus	0.060	0.055	0.050
Sulphur and phosphorus	0.110	0.105	0.100

4.2 Mechanical Properties

The bars shall meet the mechanical properties given in Table 2 when tested by the method in clause 8.1.

Table 2 Mechanical Properties

Property	DFe 415	DFe 500	DFe 550
0.2 percent proof stress, Min, N/mm ²	415	500	550
Elongation, Min, percent	14.5	12.0	10.0
Tensile strength to proof stress ratio, Min	1.10	1.08	1.06

4.3 Bend Test

The bar shall be capable of being bent through 180 degrees around a mandrel without cracks on the outer surface.

4.4 Mass Tolerance

The mass per metre of each bar shall be within plus or minus 7 percent of the nominal mass for diameters up to 10 mm.

5 PACKING

5.1 Bars shall be supplied in bundles tied with steel wire at not less than three places.

6 MARKING

6.1 Each bar shall carry a rolled-in mark identifying the manufacturer and the grade.

6.2 Each bundle shall carry a tag showing the manufacturer, grade, nominal diameter, cast number and the words "Synthetic demo label" for this illustrative document.

7 SAMPLING

7.1 One test sample shall be taken for every 100 tonnes of bars of the same diameter and grade.

7.2 If any test sample fails, two further samples shall be taken from the same lot and both shall pass.

8 TEST METHODS

8.1 Tensile Test - The tensile test shall be carried out on a bar of full cross-section as prescribed in DEMO-920.

8.2 Bend Test - The bend test shall be carried out at room temperature on a bar of full cross-section.

8.3 Chemical Analysis - The chemical composition should be determined by optical emission spectrometry.